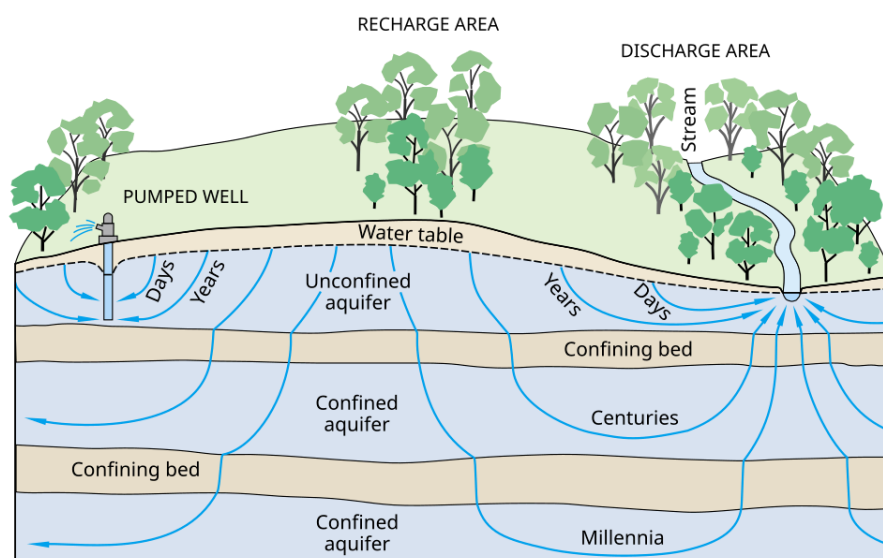


Aquifer

An **aquifer** is an underground layer of [water](#)-bearing material consisting of [permeable](#) or fractured rock, or of unconsolidated materials ([gravel](#), [sand](#), or [silt](#)).^[1] Aquifers vary greatly in their characteristics. The study of water flow in aquifers and the characterization of aquifers is called [hydrogeology](#). Related concepts include **aquitard**, a [bed](#) of low permeability along an aquifer, and **aquiclude** (or *aquifuge*), a solid and impermeable region underlying or overlying an aquifer, the pressure of which could lead to the formation of a confined aquifer. Aquifers can be classified as saturated versus unsaturated; aquifers versus aquitards; confined versus unconfined; isotropic versus anisotropic; porous, karst, or fractured; and transboundary aquifer.



Schematic of an aquifer showing confined zones, [groundwater](#) travel times, a [spring](#) and a [well](#)

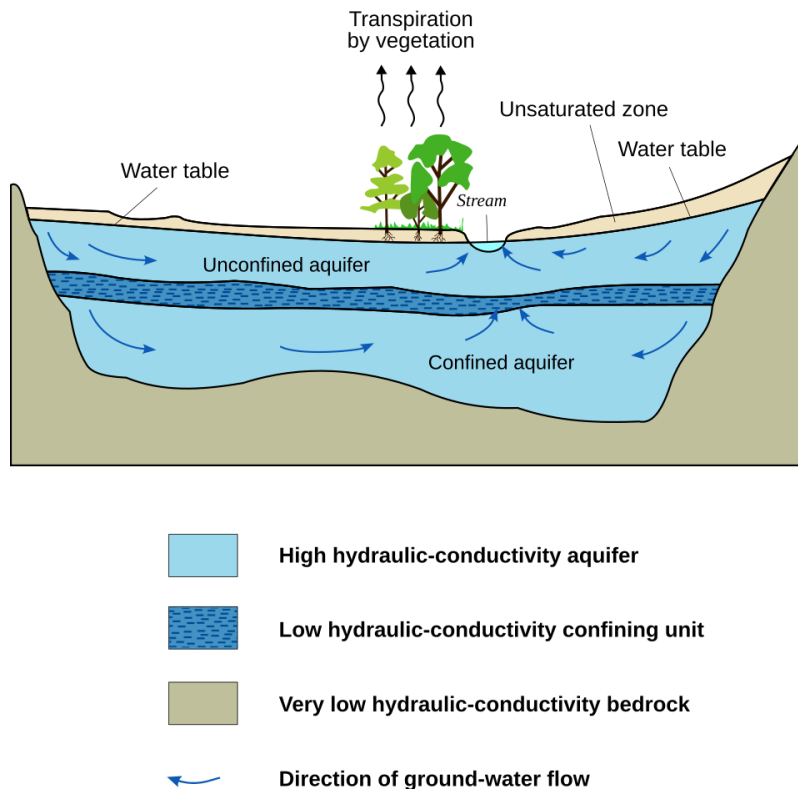
[Groundwater](#) from aquifers can be sustainably harvested by humans through the use of wells. This groundwater is mainly used for agricultural purposes but is used for other reasons such as home use, industrial use and is sometimes used as a source of [renewable energy](#).^[2] Groundwater is a major source of fresh water for many regions, although it can present various challenges for the [environment](#), such as [overdrafting](#) (extracting groundwater beyond the [equilibrium](#) yield of the aquifer), [groundwater-related subsidence](#) of land, and the [salinization](#) or [pollution](#) of the groundwater.

Etymology

The word *aquifer* is derived from the latin prefix [aqui-](#), the combining form of [aqua](#) meaning "water," and suffix [-fer](#), meaning "bearing."

Properties

Depth



An aquifer cross-section. This diagram shows two aquifers with one aquitard (a confining or impermeable layer) between them, surrounded by the bedrock *aquiclude*, which is in contact with a gaining *stream* (typical in *humid* regions). The water table and *unsaturated zone* are also illustrated.

Aquifers occur from near-surface to deeper than 9,000 metres (30,000 ft).^[3] Those closer to the surface are not only more likely to be used for water supply and irrigation, but are also more likely to be replenished by local rainfall.^[4] Although aquifers are sometimes characterized as "underground rivers or lakes," they are actually porous rock saturated with water.^[5]

Many desert areas have *limestone hills* or mountains within them or close to them that can be exploited as groundwater resources.^[6] Part of the *Atlas Mountains* in North Africa, the *Lebanon* and *Anti-Lebanon* ranges between Syria and Lebanon, the *Jebel Akhdar* in Oman, parts of the *Sierra Nevada* and neighboring ranges in the *United States' Southwest*, have shallow aquifers that are exploited for their water. *Overexploitation* can lead to the exceeding of the practical sustained yield; i.e., more water is taken out than can be replenished.

Along the coastlines of certain countries, such as [Libya](#) and [Israel](#), increased water usage associated with population growth has caused a lowering of the [water table](#) and the subsequent [contamination of the groundwater with saltwater](#) from the sea.^[7]

In 2013 large aquifers containing [offshore freshened groundwater](#) were discovered under continental shelves of Australia, China, North America and South Africa. They contain an estimated half a million cubic kilometers of "low salinity" water that could be economically processed into [potable water](#). The reserves formed when ocean levels were lower and rainwater made its way into the ground in land areas that were not submerged until the [ice age](#) ended 20,000 years ago. The volume is estimated to be 100 times the amount of water extracted from other aquifers since 1900.^{[8][9]}

Groundwater recharge

[Groundwater recharge](#) or deep drainage or deep percolation is a [hydrologic](#) process, where [water](#) moves downward from [surface water](#) to [groundwater](#). Recharge is the primary method through which water enters an aquifer. This process usually occurs in the [vadose zone](#) below plant [roots](#) and is often expressed as a [flux](#) to the [water table](#) surface. Groundwater recharge also encompasses water moving away from the water table farther into the saturated zone.^[10] Recharge occurs both naturally (through the [water cycle](#)) and through anthropogenic processes (i.e., "artificial groundwater recharge"), where rainwater and/or [reclaimed water](#) is routed to the subsurface.

Classification

An [aquitard](#) is a zone within the Earth that restricts the flow of groundwater from one aquifer to another. An aquitard can sometimes, if completely impermeable, be called an *aquiclude* or *aquifuge*. Aquitards are composed of layers of either [clay](#) or non-porous rock with low [hydraulic conductivity](#).

Saturated versus unsaturated

Groundwater can be found at nearly every point in the Earth's shallow subsurface to some degree, although aquifers do not necessarily contain [fresh water](#). The Earth's crust can be divided into two regions: the [saturated zone](#) or [phreatic zone](#) (e.g., aquifers, aquitards, etc.), where all available spaces are filled with water, and the *unsaturated zone* (also called the [vadose zone](#)), where there are still pockets of air that contain some water, but can be filled with more water.

Saturated means the pressure head of the water is greater than [atmospheric pressure](#) (it has a gauge pressure > 0). The definition of the water table is the surface where the [pressure head](#) is equal to atmospheric pressure (where gauge pressure = 0).

Unsaturated conditions occur above the water table where the pressure head is negative (absolute pressure can never be negative, but gauge pressure can) and the water that incompletely fills the pores of the aquifer material is under [suction](#). The [water content](#) in the unsaturated zone is held in place by surface [adhesive forces](#) and it rises above the water table (the zero-gauge-pressure isobar) by [capillary action](#) to saturate a small zone above the phreatic surface (the [capillary fringe](#)) at less than atmospheric pressure. This is termed tension saturation and is not the same as saturation on a water-content basis. Water content in a capillary fringe decreases with increasing distance from the phreatic surface. The capillary head depends on soil pore size. In [sandy](#) soils with larger pores, the head will be less than in clay soils with very small pores. The normal capillary rise in a clayey soil is less than 1.8 m (6 ft) but can range between 0.3 and 10 m (1 and 33 ft).^[11]

The capillary rise of water in a small-[diameter](#) tube involves the same physical process. The water table is the level to which water will rise in a large-diameter pipe (e.g., a well) that goes down into the aquifer and is open to the atmosphere.

Aquifers versus aquitards

Aquifers are typically saturated regions of the subsurface that produce an economically feasible quantity of water to a well or [spring](#) (e.g., sand and [gravel](#) or fractured [bedrock](#) often make good aquifer materials).

An aquitard is a zone within the Earth that restricts the flow of groundwater from one aquifer to another.^[12] A completely impermeable aquitard is called an *aquiclude* or *aquifuge*. Aquitards contain layers of either clay or non-porous rock with low [hydraulic conductivity](#).

In mountainous areas especially near rivers, the main aquifers are typically unconsolidated [alluvium](#), that is horizontal layers of materials deposited by water processes such as rivers and streams. In cross-section, they appear as layers of alternating coarse and fine materials. Coarse materials, because of the high energy needed to move them, tend to be found nearer to their source, such as mountain fronts or rivers, whereas the fine-grained material travels farther, to the flatter parts of the basin or overbank areas—sometimes called the pressure area. Since there are less fine-grained deposits near the source, those aquifers, also known as the fore-bay area, are often unconfined or in hydraulic communication with the land surface.

Confined versus unconfined

An unconfined aquifer has no impermeable barrier immediately above it, such that the water level can rise in response to recharge. A confined aquifer has an overlying impermeable barrier that prevents the water level in the aquifer from rising any higher. An aquifer in the same geologic unit may be confined in one area and unconfined in another. *Unconfined* aquifers are sometimes also called *water table* or *phreatic* aquifers, because their upper boundary is the [water table](#) or phreatic surface (see [Biscayne Aquifer](#)). Typically (but not always) the shallowest aquifer at a given location is unconfined, meaning it does not have a confining layer (an aquitard or aquiclude) between it and the surface. The term "perched" refers to ground water accumulating above a low-permeability unit or strata, such as a clay layer. This term is generally used to refer to a small local area of ground water that occurs at an elevation higher than a regionally extensive aquifer. The difference between perched and unconfined aquifers is their size (perched is smaller). Confined aquifers are aquifers that are overlain by a confining layer, often made up of clay. The confining layer might offer some protection from surface contamination.

If the distinction between confined and unconfined is not clear geologically (i.e., if it is not known if a clear confining layer exists, or if the geology is more complex, e.g., a fractured bedrock aquifer), the value of storativity returned from an [aquifer test](#) can be used to determine it (although aquifer tests in unconfined aquifers should be interpreted differently than confined ones). Confined aquifers have very low [storativity](#) values (much less than 0.01, and as little as 10^{-5}), which means that the aquifer is storing water using the mechanisms of aquifer matrix expansion and the compressibility of water, which typically are both quite small quantities. Unconfined aquifers have storativities (typically called [specific yield](#)) greater than 0.01 (1% of bulk volume); they release water from storage by the mechanism of actually draining the pores of the aquifer, releasing relatively large amounts of water (up to the drainable [porosity](#) of the aquifer material, or the minimum volumetric [water content](#)).

Isotropic versus anisotropic

In [isotropic](#) aquifers or aquifer layers the hydraulic conductivity (K) is equal for flow in all directions, while in [anisotropic](#) conditions it differs, notably in horizontal (K_h) and vertical (K_v) sense.

Semi-confined aquifers with one or more aquitards work as an anisotropic system, even when the separate layers are isotropic, because the compound K_h and K_v values are different (see [hydraulic transmissivity](#) and [hydraulic resistance](#)).

When calculating [flow to drains](#)^[13] or [flow to wells](#)^[14] in an aquifer, the anisotropy is to be taken into account lest the resulting design of the drainage system may be faulty.

Porous, karst, or fractured

To properly manage an aquifer its properties must be understood. Many properties must be known to predict how an aquifer will respond to rainfall, drought, pumping, and [contamination](#).

Considerations include where and how much water enters the groundwater from rainfall and snowmelt, how fast and in what direction the groundwater travels, and how much water leaves the ground as springs. [Computer models](#) can be used to test how accurately the understanding of the aquifer properties matches the actual aquifer performance.^{[15]:192–193,233–237} Environmental regulations require sites with potential sources of contamination to demonstrate that the [hydrology](#) has been [characterized](#).^{[15]:3}

Porous



Water in porous aquifers slowly seeps through pore spaces between sand grains

Porous aquifers typically occur in sand and [sandstone](#).^[16] Their properties depend on the [depositional sedimentary environment](#) and later natural cementation of the sand grains. The environment where a sand body was deposited controls the orientation of the sand grains, the horizontal and vertical variations, and the distribution of shale layers. Even thin shale layers are important barriers to groundwater flow. All these factors affect the [porosity](#) and [permeability](#) of sandy aquifers.^{[17]:413}

Sandy deposits formed in [shallow marine environments](#) and in [windblown sand dune environments](#) have moderate to high permeability while sandy deposits formed in [river environments](#) have low to moderate permeability.^{[17]:418} Rainfall and snowmelt enter the groundwater where the aquifer is near the surface. Groundwater flow directions can be determined from [potentiometric surface](#) maps of water levels in wells and springs. [Aquifer tests](#) and [well tests](#) can be used with [Darcy's law](#) flow equations to determine the ability of a porous aquifer to convey water.^{[15]:177–184}

Analyzing this type of information over an area gives an indication how much water can be pumped without [overdrafting](#) and how contamination will travel.^{[15]:233} In porous aquifers groundwater flows as slow seepage in pores between sand grains. A groundwater flow rate of 1 foot per day (0.3 m/d) is considered to be a high rate for porous aquifers,^[18] as illustrated by the water slowly seeping from sandstone in the accompanying image to the left.

Porosity is important, but, *alone*, it does not determine a rock's ability to act as an aquifer. Areas of the [Deccan Traps](#), a [basaltic](#) lava formation in west-central India, are examples of rock formations with high porosity but low permeability, making them poor aquifers. Similarly, the micro-porous (Upper [Cretaceous](#)) [Chalk Group](#) of south east England, although having a reasonably high porosity, has a low grain-to-grain permeability, with its good water-yielding characteristics mostly due to micro-fracturing and fissuring.

Karst



Water in karst aquifers can form [subterranean rivers](#).

[Karst](#) aquifers typically develop in [limestone](#). Surface water containing natural [carbonic acid](#) moves down into small fissures in limestone. This carbonic acid gradually dissolves limestone thereby enlarging the fissures. The enlarged fissures allow a larger quantity of water to enter which leads to a progressive enlargement of openings. Abundant small openings store a large quantity of water. The larger openings form a conduit system that drains the aquifer to springs.^[19]

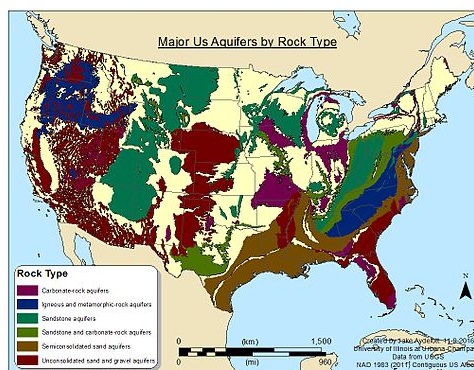
Characterization of karst aquifers requires field exploration to locate [sinkholes](#), [swallets](#), [sinking streams](#), and [springs](#) in addition to studying [geological maps](#).^{[20]:4} Conventional hydrogeologic methods such as aquifer tests and potentiometric mapping are insufficient to characterize the complexity of karst aquifers. These conventional investigation methods need to be supplemented with [dye traces](#), measurement of spring discharges, and analysis of water chemistry.^[21] U.S. Geological Survey dye tracing has determined that conventional groundwater models that assume a uniform distribution of porosity are not applicable for karst aquifers.^[22]

Linear alignment of surface features such as straight stream segments and sinkholes develop along [fracture traces](#). Locating a well in a fracture trace or intersection of fracture traces increases the likelihood to encounter good water production.^[23] Voids in karst aquifers can be large enough to cause destructive collapse or [subsidence](#) of the ground surface that can initiate a catastrophic release of contaminants.^{[15]:3–4} Groundwater flow rate in karst aquifers is much more rapid than in porous aquifers as shown in the accompanying image to the left. For example, in the Barton Springs Edwards aquifer, dye traces measured the karst groundwater flow rates from 0.5 to 7 miles per day (0.8 to 11.3 km/d).^[24] The rapid groundwater flow rates make [karst aquifers much more sensitive](#) to groundwater contamination than porous aquifers.^{[20]:1}

In the extreme case, groundwater may exist in *underground rivers* (e.g., [caves](#) underlying [karst topography](#)).

Fractured

If a rock unit of low [porosity](#) is highly fractured, it can also make a good aquifer (via [fissure](#) flow), provided the rock has a hydraulic conductivity sufficient to facilitate movement of water.



Map of major US aquifers by rock type

Undersea

Remote sensing in 2015 and direct sampling in 2025 discovered an undersea reservoir of fresh water off the coast of [Nantucket](#), stretching possibly from New Jersey to Maine and thus representing one of the largest freshwater aquifers in the United States. The aquifer itself may represent glacial meltwater once overlain by dry land that subsequently was flooded by rising [sea levels](#)—as caused by the [Holocene glacial retreat](#). Or it, (the aquifer) may maintain long-term groundwater flow connections along the US east coast, and thereby is recharged even today by contemporary rainfall over coastal surface lands.^[25]

Human use of groundwater

Reliance on groundwater will only increase, mainly due to growing water demand by all sectors combined with [increasing variation in rainfall patterns](#).^[26] Safe use of groundwater varies substantially by the elements present and use-cases, with significant differences between consumption for humans, livestock and different crops.^[27]

Challenges for using groundwater include: [overdrafting](#) (extracting groundwater beyond the [equilibrium](#) yield of the aquifer), [groundwater-related subsidence](#) of land, groundwater becoming saline, and [groundwater pollution](#).

By country or continent



[Texas blind salamander](#) found in [Edwards Aquifer](#)

Africa

Aquifer depletion is a problem in some areas, especially in northern [Africa](#), where one example is the [Great Manmade River](#) project of [Libya](#). However, new methods of groundwater management such as artificial recharge and injection of surface waters during seasonal wet periods has extended the life of many freshwater aquifers, especially in the United States.

Australia

The [Great Artesian Basin](#) situated in [Australia](#) is arguably the largest groundwater aquifer in the world (over 1.7 million km² or 0.66 million sq mi).^[28] It plays a large part in water supplies for [Queensland](#), some remote parts of [South Australia](#) and most of the [Northern Territory](#).^[29]

Canada

Discontinuous sand bodies at the base of the [McMurray Formation](#) in the [Athabasca Oil Sands](#) region of northeastern [Alberta](#), Canada, are commonly referred to as the [Basal Water Sand \(BWS\) aquifers](#).^[30] Saturated with water, they are confined beneath impermeable [bitumen](#)-saturated sands that are exploited to recover bitumen for [synthetic crude](#) oil production. Where they are deep-lying and recharge occurs from underlying [Devonian formations](#) they are saline, and where they are shallow and recharged by [surface water](#) they are non-saline. The BWS typically pose problems for the recovery of bitumen, whether by [open-pit mining](#) or by *in situ* methods such as [steam-assisted gravity drainage](#) (SAGD), and in some areas they are targets for waste-water injection.^{[31][32][33]}

South America

The [Guarani Aquifer](#), located beneath the surface of [Argentina](#), [Brazil](#), [Paraguay](#), and [Uruguay](#), is one of the world's largest aquifer systems and is an important source of [fresh water](#).^[34] Named after the [Guarani people](#), it covers 1,200,000 km² (460,000 sq mi), with a volume of about 40,000 km³ (9,600 cu mi), a thickness of between 50 and 800 m (160 and 2,620 ft) and a maximum depth of about 1,800 m (5,900 ft).

United States

The [Ogallala Aquifer](#) of the central United States is one of the world's great aquifers, but in places it is being rapidly [depleted](#) by growing municipal use, and continuing agricultural use.^[35] This huge aquifer, which underlies portions of eight states, contains primarily [fossil water](#) from the time of the last [glaciation](#). Annual recharge, in the more arid parts of the aquifer, is estimated to total only about 10 percent of annual withdrawals. According to a 2013 report by the [United States Geological Survey](#) (USGS), the depletion between 2001 and 2008, inclusive, is about 32 percent of the cumulative depletion during the entire 20th century.^[36]

In the United States, the biggest users of water from aquifers include agricultural irrigation and oil and coal extraction.^[37] "Cumulative total [groundwater depletion](#) in the United States accelerated in the late 1940s and continued at an almost steady linear rate through the end of the century. In addition to widely recognized environmental consequences, groundwater depletion also adversely impacts the long-term sustainability of groundwater supplies to help meet the Nation's water needs."^[36]

An example of a significant and sustainable carbonate aquifer is the [Edwards Aquifer](#)^[38] in central [Texas](#). This carbonate aquifer has historically been providing high quality water for nearly 2 million people, and even today, is full because of tremendous recharge from a number of area streams, rivers and [lakes](#). The primary risk to this resource is human development over the recharge areas.

See also

- [Aquifer storage and recovery](#) – Injection of water into an aquifer for later use
- [Aquifer Susceptibility](#) – Ability of permeable rock to hold liquids
- [Artesian aquifer](#) – Naturally-pressurized water source
- [Catchment hydrology](#) – Hydrology of drainage basins
- [Hydraulic tomography](#) – Use of multiple hydraulic tests to characterize an aquifer
- [Overexploitation](#) – Depleting a renewable resource
- [Seasonal thermal energy storage](#) – Storage of heat or cold for periods of up to several months
- [Spring \(hydrology\)](#) – Point at which water emerges from an aquifer to the surface
- [Surficial aquifer](#) – Shallow, thin aquifer

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External links

- IGRAC International Groundwater Resources Assessment Centre (<https://www.un-igrac.org/>)
- The Groundwater Project (<https://gw-project.org/>) - Online platform for groundwater knowledge